

Finding Inverses with Domain Restrictions

Inverting a measured model, restricting a domain before inverting, and checking that two rules undo each other

Directions

- To invert a rule, swap the roles of input and output and solve for the new output.
- Every applied problem states its units. Answer in the unit printed on the answer line.
- A rule that is not one-to-one has no inverse until its domain is cut down — say what the cut is before you take a square root.

1. A phone plan charges $C(m) = 0.05m + 20$, where C is the monthly cost in dollars and m is the number of call minutes used. A bill came to 45 dollars. Use the inverse of C to find the number of minutes used.

Answer: _____ min

2. Let $f(x) = 4x + 9$ and $g(x) = \frac{x-9}{4}$. Compute $(f \circ g)(x)$ and simplify it to show that g undoes f .

Answer: _____

3. The area of a circle is $A(r) = \pi r^2$, where r is the radius in centimetres and A is the area in square centimetres. Only $r > 0$ is meaningful.

(a) Write the domain of A^{-1} as an inequality in A .

Answer: _____

(b) A circle has area 64π square centimetres. Use A^{-1} to find its radius.

Answer: _____ cm

4. A spring hangs from a hook. With a load of w kilograms attached, its length is $L(w) = 12 + 0.4w$ centimetres. Find the load that stretches the spring to a length of 22 centimetres.

Answer: _____ kg

5. Let $f(x) = \frac{2x+1}{x-3}$ and $g(x) = \frac{3x+1}{x-2}$. Compute $(f \circ g)(x)$ and simplify it to show that g undoes f .

Answer: _____

6. A stone dropped from rest falls $d(t) = 5t^2$ metres in t seconds, for $t \geq 0$.

(a) The stone has fallen 80 metres. Use the inverse of d to find the time taken.

Answer: _____ s

(b) Explain why the restriction $t \geq 0$ is what allows d to have an inverse at all.

Answer: _____

7. Temperature converts by $F(c) = \frac{9}{5}c + 32$, where c is in degrees Celsius and F is in degrees Fahrenheit. Find the Celsius temperature that corresponds to 5 degrees Fahrenheit.

Answer: _____ degrees Celsius

8. Let $f(x) = x^2 - 6x + 5$ with the domain restricted to $x \geq 3$.

(a) Complete the square, then write the domain of f^{-1} as an inequality in x .

Answer: _____

(b) Find $f^{-1}(21)$.

Answer: _____

9. Let $f(x) = \frac{1}{x-4}$ and $g(x) = \frac{1}{x} + 4$. Compute $(f \circ g)(x)$ and simplify it to show that g undoes f .

Answer: _____

10. A ball is thrown straight down from a 45 metre ledge. Its height is $h(t) = 45 - 5t^2$ metres, for $0 \leq t \leq 3$ seconds.

- (a) Find the time at which the ball is 25 metres above the ground.

Answer: _____ s

- (b) Explain why the restriction $0 \leq t \leq 3$ has to be stated before this model can be inverted.

Answer: _____